June 8, 2024

[3]:

```python
import numpy as np
import matplotlib.pyplot as plt
from scipy.stats import binom
```

[4]:

```python
p1 = 0.05  #
p2 = 0.05  #
x0 = 20    #
y0 = 20    #
x_range = np.arange(10, 101)  #
y_range = np.arange(10, 101)  #
```

[5]:

```python
#
def indiff_curve(x_range, y_range, p, x0):
    curve = []
    for x in x_range:
        min_y = None
        for y in y_range:
            prob = sum(binom.pmf(k, x, 1-(1-p)**y) for k in range(x0, x+1))
            if prob >= 0.9:
                min_y = y
                break
        if min_y is not None:
            curve.append((x, min_y))
    return np.array(curve)
```

[11]:

```python
#
indiff_curve_a = indiff_curve(x_range, x_range, p1, x0)
indiff_curve_b = indiff_curve(x_range, x_range, p2, y0)

#
min_y_for_a = np.interp(x_range, indiff_curve_a[:, 0], indiff_curve_a[:, 1],␣
 ↪left=100, right=100)
min_y_for_b = np.interp(x_range, indiff_curve_b[:, 0], indiff_curve_b[:, 1],␣
 ↪left=100, right=100)
common_safe_y = np.maximum(min_y_for_a, min_y_for_b)
```

[12]:

```
#
plt.figure(figsize=(10, 8))
plt.plot(indiff_curve_a[:, 0], indiff_curve_a[:, 1], label='Indifference Curve␣
    ↪for A (x >= f(y))')
plt.plot(indiff_curve_b[:, 1], indiff_curve_b[:, 0], label='Indifference Curve␣
    ↪for B (y >= g(x))')

#
plt.fill_between(x_range, common_safe_y, 100, color='blue', alpha=0.1,␣
    ↪label='Common Safe Zone')

plt.xlabel('Number of Nukes for A (x)')
plt.ylabel('Number of Nukes for B (y)')
plt.title('Indifference Curves and Common Safe Zone for Nuclear Arms Race')
plt.legend()
plt.grid(True)
plt.show()
```